

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0643

Stumpy Lake 138 kV
March 25, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0643. The Interconnection Customer proposes a 298.4 MW battery energy storage facility interconnecting at the Stumpy Lake 138 kV substation in the Virginia area. The request was received on July 12, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 245 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$46,297,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Stumpy Lake to F B Culley 138.0 kV	Victory Way to Maine Yankee 138.0 kV	191.0	215	100.0
Victory Way to Maie Yankee 138.0 kV	Stumpy Lake to F B Culley 138.0 kV	181.5	226	100.0
Green Run to Stumpy Lake 138.0 kV	F B Culley to Stumpy Lake 138.0 kV	180.8	186	100.0
Maine Yankee to Green Run Substation 138.0 kV	Stumpy Lake to F B Culley 138.0 kV	179.0	204	100.0
Copell to F B Culley 138.0 kV	Moval Substation to Ottawa Avenue 138.0 kV	135.3	251	39.2
Travis Ab to M.L. Hibbard S.E. Station 138.0 kV	Mccartney to Yarnen 345.0 kV	124.2	304	7.8
Elizabeth Riv Power Station to Chesapeake Energy Center 138.0 kV Line	Yadkin to Harborgen 138.0 kV	124.1	200	22.2
Avra Valley to Windsor Dp 138.0 kV	Yarnen to Mira Vsta 345.0 kV Line	123.0	343	9.8
Moval to Ottawa Avenue 138.0 kV L	Coppell to F B Culley 138.0 kV	121.1	272	31.5
Werner W to Windsor Dp 138.0 kV	Yarnen to Mira Vista 345.0 kV line	118.7	363	9.8
Mcindoes to Coppell 138.0 kV #1	Mcindoes to Coppell 138.0 kV Line 2	114.1	283	21.0
Thalia to Igo 138.0 kV #1	Thalia Substation to Igo 138.0 kV Ckt 2	108.9	291	14.4
Wheelabrator Portsmouth to Victory Way 138.0 kV	Stumpy Lake to F B Culley 138.0 kV	108.3	234	30.6
Sully Buttes to Yorktown 138.0 kV L	Newport News 345/1 kV Transformer	106.2	318	7.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Moal to F B Culley 138.0 kV Ckt 1	F B Culley Substation to Moal 138.0 kV Cir. 2	105.6	275	28.3
F B Culley Substation to Moal 138.0 kV Cir. 2	Moal to F B Culley 138.0 kV Circuit 1	105.6	275	28.3
Travis Ab to Werner West Substation 138.0 kV	Mccartney Substation to Yarnen 345.0 kV line	104.5	492	8.4
Allegheny Energy Units 3 4 & 5 345/1kV Autotransformer	Hilop to Piney Creek Project 500.0 kV	104.0	1,302	6.3
Allegheny Energy Units 3 4 & 5 500/1kV TX	Hilltop to Piney Creek Project 500.0 kV	104.0	1,302	6.3
Coppell to Mcindoes 138.0 kV Circuit 2	Coppell to Mcindoes 138.0 kV	104.0	310	21.0
Hartselle 345-138 kV Auto	Mira Vista to Yarnen 345.0 kV	103.5	569	14.9
Sugar Hill to Allegheny Energy Units 3 4 & 5 345.0 kV	Piney Crk Project to Hilltp 500.0 kV	103.4	1,438	6.4
Warrenton to Oakwood 138.0 kV	Yadkin to Mickns Substation 138.0 kV	103.2	272	15.2
Fyffe 345 kV/138 kV Auto	Yarnen to Mccartney 345.0 kV	102.2	881	26.7
Avra Valley to Chuckatuck 138.0 kV L	Mira Vista to Yarnen 345.0 kV Line	101.9	527	19.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Stumpy Lake to F B Culley 138.0 kV	Rebuild line with higher-capacity conductor	\$1,673,000	100% (\$1,673,000)
Victory Way to Maine Yankee 138.0 kV	Rebuild line with higher-capacity conductor	\$1,135,000	100% (\$1,135,000)
Stumpy Lake to Green Run 138.0 kV	Rebuild line with higher-capacity conductor	\$1,406,000	100% (\$1,406,000)
Maine Yankee to Green Run Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$6,818,000	100% (\$6,818,000)
Copell to F B Culley 138.0 kV	Rebuild line with higher-capacity conductor	\$2,042,000	100% (\$2,042,000)
Travis Ab to M.L. Hibbard S.E. Sta 138.0 kV	Rebuild line with higher-capacity conductor	\$31,636,000	15% (\$4,746,000)
Elizabeth Riv Power Sta to Chesapeake Energy Ctr Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$724,000	100% (\$724,000)
Avra Valley to Windsor Dp 138.0 kV	Rebuild line with higher-capacity conductor	\$3,084,000	14% (\$441,000)
Moal to Ottawa Avenue 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,386,000	100% (\$2,386,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Werner W to Windsor Dp 138.0 kV	Rebuild line with higher-capacity conductor	\$8,531,000	14% (\$1,219,000)
Mcindoes to Coppel 138.0 kV	Rebuild line with higher-capacity conductor	\$8,950,000	100% (\$8,950,000)
Thalia to Igo 138.0 kV #1	Reconductor line	\$2,038,000	51% (\$1,040,000)
Victory Way to Wheelabrator Portsmouth 138.0 kV Line	Reconductor line	\$2,768,000	65% (\$1,800,000)
Sully Buttes to Yorktown 138.0 kV L	Reconductor line	\$453,000	41% (\$185,000)
Moval to F B Culley 138.0 kV Ckt 1	Reconductor line	\$500,000	100% (\$500,000)
F B Culley Substation to Moval 138.0 kV Cir. 2	Reconductor line	\$631,000	100% (\$631,000)
Travis Ab to Werner West Substation 138.0 kV	Reconductor line	\$3,454,000	7% (\$256,000)
Allegheny Energy Units 3 4 & 5 345/1kV Autotransformer	Replace transformer with higher-rated unit	\$16,532,000	3% (\$560,000)
Allegheny Energy Units 3 4 & 5 500-1 kV Auto	Replace transformer with higher-rated unit	\$21,386,000	3% (\$724,000)
Coppel to Mcindoes 138.0 kV Circuit 2	Reconductor line	\$2,938,000	100% (\$2,938,000)
Hartselle 345-138 kV Auto	Replace transformer with higher-rated unit	\$20,691,000	17% (\$3,553,000)
Sugar Hill to Allegheny Energy Units 3 4 & 5 345.0 kV L	Reconductor line	\$6,429,000	3% (\$218,000)
Warrenton to Oakwood 138.0 kV	Reconductor line	\$720,000	100% (\$720,000)
Fyffe 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$10,545,000	13% (\$1,350,000)
Avra Valley to Chuckatuck 138.0 kV L	Reconductor line	\$1,917,000	15% (\$282,000)
	Total allocated to project		\$46,297,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.